Vortex Lattice Method (V.L.M.)

Aerodynamic Characterization and Stall Optimization

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1. Objectives of the Project

The objectives of this project are the following:

- Apply the vortex lattice method to determine the aerodynamic characteristics of a wing.
- Redesign the wing so that its stall starts near the root.

2. Baseline Data

The baseline airfoil is the NACA 3217, which has, relative to the chord, a maximum camber f of 3 %, located at $x_f = 0.20\,c$, and a maximum thickness t of 17 % (Figure 1).

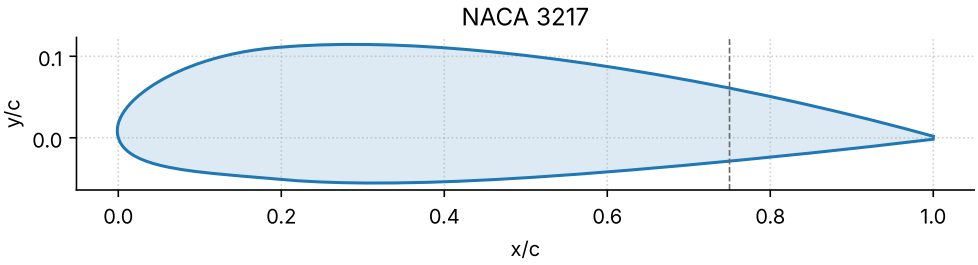


Figure 1: NACA 3217 airfoil. The dashed line marks the control point at $x/c = 0.75$.

The wing parameters follow from the assignment values $W = 9$, $X = 3$, $Y = 5$, $Z = 5$, and are listed in Table 1.

Geometric parameter	Value
Wing area (S_w)	$1\,\mathrm{m}^2$
Aspect ratio (A)	15
Taper ratio (λ)	0.65
Quarter-chord sweep ($\Lambda_{1/4}$)	12°
Geometric twist at the tip (ε), linear washout	-5°
Chordwise panels (N_x)	1
Spanwise panels per semispan (N_y)	10

Table 1: Geometric parameters of the wing.

2.1. Wing Representation

Figure 2 shows the planform and the lattice: one horseshoe vortex per panel, bound on the quarter-chord line, with its control point at three quarters of the local chord.

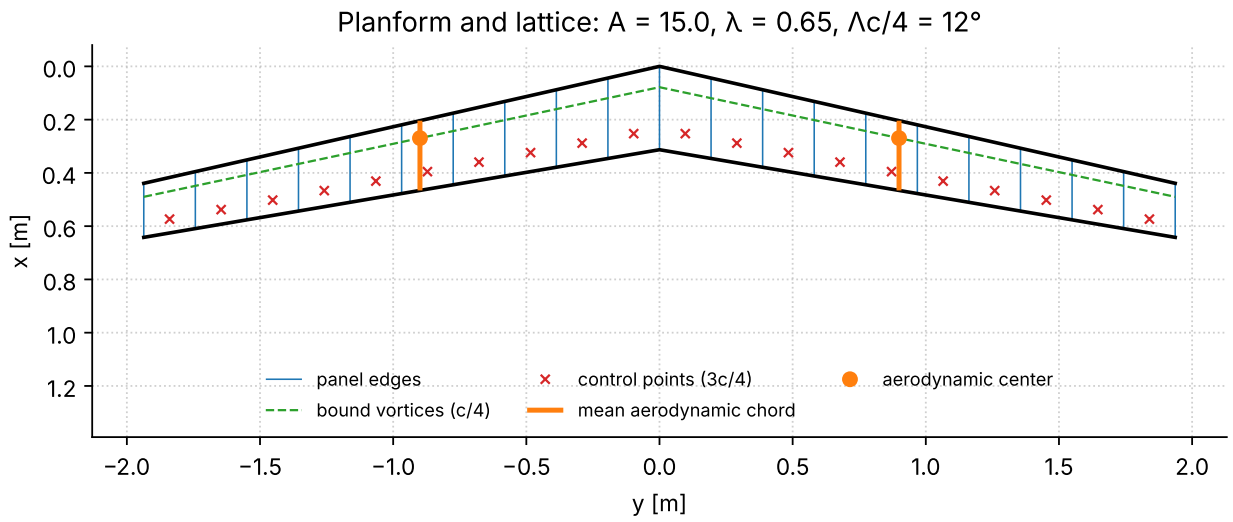


Figure 2: Wing planform, vortex lattice, mean aerodynamic chord and aerodynamic center.

3. Calculation of Geometric Parameters

From the data of the statement, the following parameters are obtained [1].

- **Camber-line slope at the control point.** For a NACA four-digit airfoil, aft of the maximum camber ($x/c = 0.75 > x_f = 0.20$), and with $f = 0.03$, $x_f = 0.20$:

$$\frac{\mathrm{d}z_c}{\mathrm{d}x} = \frac{2f}{(1-x_f)^2}\left(x_f - \frac{x}{c}\right) \tag{1}$$

- **Wingspan:**

$$A = \frac{b^2}{S} \tag{2}$$

- **Root chord**, solving

$$S = \frac{b}{2}\, c_0\,(1+\lambda) \tag{3}$$

- **Tip chord:**

$$c_t = \lambda\, c_0 \tag{4}$$

- **Geometric mean chord:**

$$c_{gm} = \frac{c_0 + c_t}{2} \tag{5}$$

- **Mean aerodynamic chord:**

$$c_{am} = \frac{2}{3}\, c_0\, \frac{1+\lambda+\lambda^2}{1+\lambda} \tag{6}$$

- **Aerodynamic center of the wing.** The leading-edge sweep follows from the quarter-chord sweep and the taper,

$$\tan\Lambda_{LE} = \tan\Lambda_{1/4} + \frac{c_0}{2b}\,(1-\lambda) \tag{7}$$

and the aerodynamic center lies at a quarter of the mean aerodynamic chord, at its spanwise station:

$$y_{ca} = \frac{b}{6}\,\frac{1+2\lambda}{1+\lambda} \tag{8}$$

$$x_{ca} = y_{ca}\tan\Lambda_{LE} + 0.25\,c_{am} \tag{9}$$

The resulting values are given in Table 2.

Geometric parameter	Value
Camber-line slope at $x = 3c/4$	-0.0516
Wingspan (b)	3.873 m
Root chord (c_0)	0.313 m
Tip chord (c_t)	0.203 m
Geometric mean chord (c_{gm})	0.258 m
Mean aerodynamic chord (c_{am})	0.262 m
Leading-edge sweep (Λ_{LE})	12.78°
Aerodynamic center (x_{ca}, y_{ca})	(0.2694, 0.8994) m

Table 2: Derived wing parameters.

4. Local Lift Coefficients

For each angle of attack α, the spanwise circulation $\{\Gamma_j\}$ is found by solving the linear system

$$[a_{ij}]\left\{\frac{\Gamma_j}{U_\infty}\right\} = -\{b_i\}, \qquad b_i = \alpha + \varepsilon(y_i) - \left(\frac{\mathrm{d}z_c}{\mathrm{d}x}\right)_{3c/4} \tag{10}$$

where $[a_{ij}]$ is the influence matrix of the horseshoe-vortex lattice (normal velocity induced at control point i by a unit vortex on panel j, from the Biot–Savart law [2]), and b_i expresses the no-penetration condition at each control point, including the local geometric twist $\varepsilon(y_i)$ and the camber-line slope of Equation (1).

The Kutta–Joukowski theorem then gives the sectional lift coefficient:

$$c_l(y_i) = \frac{2\Gamma(y_i)}{U_\infty\, c(y_i)} \tag{11}$$

Table 3 lists the results for the right half-wing (the wing is symmetric) and Figure 3 shows the whole span.

Panel	y [m]	$c_l(-12°)$	$c_l(-6°)$	$c_l(0°)$	$c_l(6°)$	$c_l(12°)$
1	0.097	-0.881	-0.342	0.198	0.737	1.277
2	0.290	-0.939	-0.379	0.181	0.741	1.302
3	0.484	-0.993	-0.420	0.154	0.727	1.300
4	0.678	-1.044	-0.462	0.119	0.701	1.283
5	0.871	-1.092	-0.505	0.081	0.667	1.254
6	1.065	-1.135	-0.547	0.040	0.628	1.216
7	1.259	-1.169	-0.585	-0.001	0.583	1.167
8	1.452	-1.186	-0.614	-0.041	0.531	1.103
9	1.646	-1.161	-0.619	-0.077	0.465	1.007
10	1.840	-0.997	-0.546	-0.094	0.358	0.810

Table 3: Sectional lift coefficient of each panel (right half-wing) for each angle of attack.

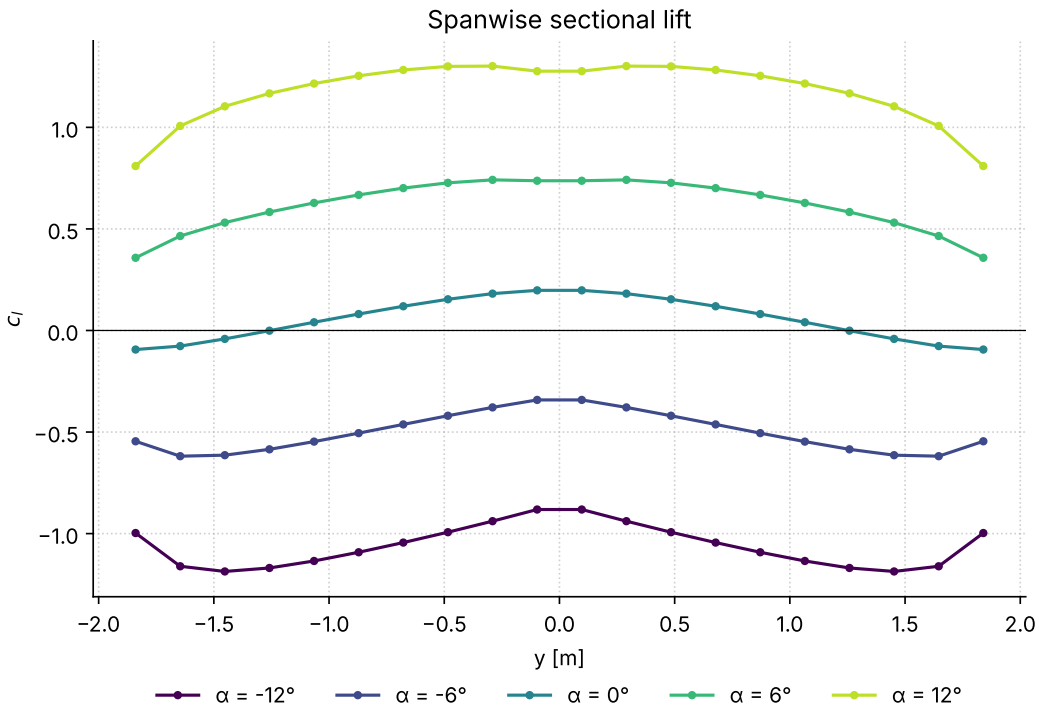


Figure 3: Spanwise sectional lift distribution. The dip at the root is a sweep effect.

5. Global Lift Coefficient

The global lift coefficient is obtained by integrating the sectional lift along the span:

$$C_L = \frac{1}{S} \sum_{i=1}^{N} c_{l,i}\, c_i\, \Delta y_i \tag{12}$$

The model is linear (attached, inviscid flow), so $C_L(\alpha)$ is a straight line. Its slope and its value at $\alpha = 0$ are obtained by a least-squares fit through the five angles of attack,

$$C_L = C_{L,\alpha=0} + C_{L\alpha}\, \alpha \tag{13}$$

and the zero-lift angle follows as

$$\alpha_{L=0} = -\frac{C_{L,\alpha=0}}{C_{L\alpha}} \tag{14}$$

which measures the combined effect of camber and twist on the complete wing.

$C_{L\alpha}$	$C_{L,\alpha=0}$	$\alpha_{L=0}$
$0.0932\,\mathrm{deg}^{-1}$ $(5.342\,\mathrm{rad}^{-1})$	0.068	$-0.733°$

Table 4: Lift-curve slope, lift at zero incidence and zero-lift angle.

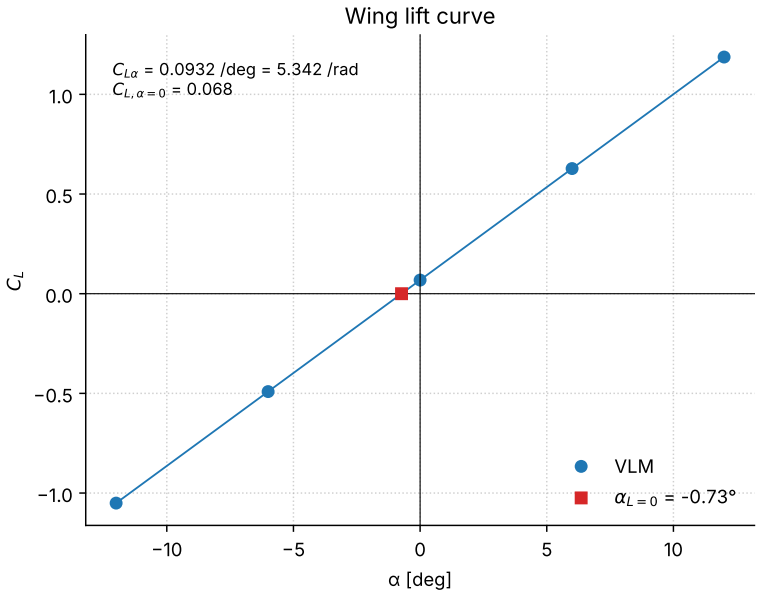


Figure 4: Wing lift curve $C_L(\alpha)$.

Figure 4 shows the linear response with $C_{L,\alpha=0} = 0.068$: the camber of the airfoil produces lift at zero incidence, partly offset by the washout of the tips.

6. Basic and Unit Additional Lift Distribution

Because the problem is linear, the sectional lift can be split into a basic part, due to camber and twist, and a unit additional part, due to the angle of attack and scaled by the global lift:

$$c_l(y) = c_{l,b}(y) + c_{l,a}(y)\, C_L \tag{15}$$

- **Basic lift** $c_{l,b}$: the system of Equation (10) is solved at the zero-lift angle. The global coefficient is zero, but twist and camber still produce a non-zero local load:

$$[a_{ij}] \left\{ \frac{\Gamma_{b,j}}{U_\infty} \right\} = -\left(\alpha_{L=0} + \varepsilon(y_i) - \frac{\mathrm{d}z_c}{\mathrm{d}x} \right), \qquad c_{l,b} = \frac{2\Gamma_b}{U_\infty c} \tag{16}$$

- **Unit additional lift** $c_{l,a}$: twist and camber are removed, a uniform incidence is applied, and the result is normalized to $C_L = 1$:

$$c_{l,a} = \frac{c_{l,\text{planform}}(y)}{C_{L,\text{planform}}} \tag{17}$$

Panel	y [m]	$c_{l,b}$	$c_{l,a}$
1	0.097	0.132	0.964
2	0.290	0.113	1.001
3	0.484	0.084	1.025
4	0.678	0.048	1.040
5	0.871	0.009	1.048
6	1.065	−0.031	1.050
7	1.259	−0.072	1.044
8	1.452	−0.111	1.023
9	1.646	−0.143	0.969
10	1.840	−0.149	0.807

Table 5: Basic and unit additional lift coefficients (right half-wing).

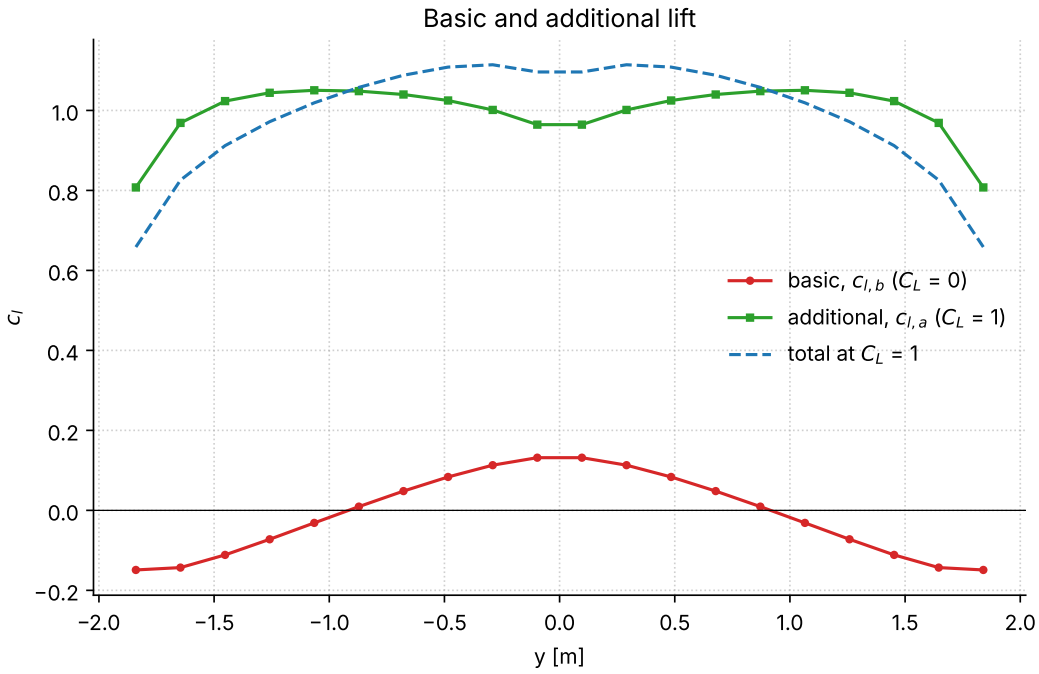


Figure 5: Basic and unit additional lift distributions.

The basic load is positive inboard and negative at the tips: the washout unloads the tips. The additional load is fairly flat over most of the span, with the root dip caused by sweep, and falls off near the tips.

7. Induced Drag Coefficients

The downwash produced by the trailing vortices is evaluated in the Trefftz plane, far downstream. The influence matrix of the semi-infinite trailing legs is

$$A^T_{ij} = \frac{1}{2\pi}\left(\frac{1}{y_i - y_{\text{left},j}} - \frac{1}{y_i - y_{\text{right},j}}\right) \tag{18}$$

Multiplying it by the circulation vector gives the far-wake normal velocity w_T. The induced angle at the wing is half of it,

$$\alpha_i = \frac{w_T}{2U_\infty} \tag{19}$$

and the local induced drag is the projection of the local lift onto the inclined flow:

$$c_{di}(y) = c_l(y)\,\alpha_i(y) \tag{20}$$

The global coefficient is obtained by integration over the span:

$$C_{Di} = \frac{1}{S}\sum_{i=1}^{N} c_{di,i}\,c_i\,\Delta y_i \tag{21}$$

Panel	y [m]	$c_{di}(-12^\circ)$	$c_{di}(-6^\circ)$	$c_{di}(0^\circ)$	$c_{di}(6^\circ)$	$c_{di}(12^\circ)$
1	0.097	0.0061	−0.0006	0.0021	0.0143	0.0360
2	0.290	0.0114	0.0008	0.0014	0.0132	0.0362
3	0.484	0.0144	0.0020	0.0008	0.0108	0.0321
4	0.678	0.0170	0.0031	0.0003	0.0087	0.0281
5	0.871	0.0199	0.0044	0.0001	0.0069	0.0249
6	1.065	0.0235	0.0060	0.0000	0.0055	0.0226
7	1.259	0.0286	0.0080	0.0000	0.0046	0.0218
8	1.452	0.0363	0.0109	0.0002	0.0041	0.0227
9	1.646	0.0487	0.0154	0.0006	0.0044	0.0266
10	1.840	0.0657	0.0213	0.0012	0.0052	0.0335

Table 6: Local induced drag coefficient (right half-wing).

α	-12°	-6°	0°	6°	12°
C_{Di}	0.0251	0.0063	0.0007	0.0082	0.0288

Table 7: Global induced drag coefficient.

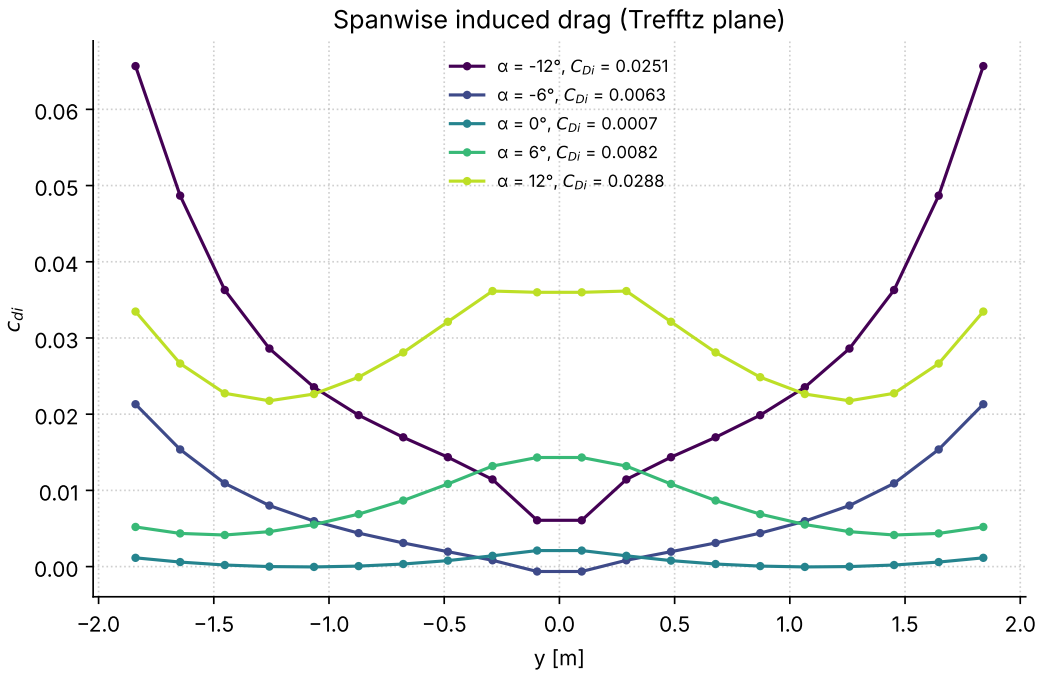


Figure 6: Spanwise induced drag distribution.

Induced drag is smallest near zero incidence and grows with $|\alpha|$. At negative angles it concentrates at the tips, where the washout adds to the negative load. The small negative local values near the root at -6° are local upwash, not a net thrust: the global C_{Di} is positive at every angle.

8. Pitching Moment

The pitching moment is computed about two reference points: the coordinate origin (root leading edge) and the aerodynamic center of the wing, x_{ca}. In this one-chordwise-panel model the lift of each panel acts on its quarter-chord line:

$$x_{1/4}(y) = |y| \tan \Lambda_{LE} + 0.25\, c(y) \tag{22}$$

The moment about the origin (nose-up positive, reference chord c_{am}) is

$$C_{M,0y} = \frac{1}{S\, c_{am}} \sum_{i=1}^{N} \left(-c_{l,i}\, c_i\, x_{1/4,i}\, \Delta y_i \right) \tag{23}$$

and it is transferred to the aerodynamic center with

$$C_{M,ca} = C_{M,0y} + C_L \frac{x_{ca}}{c_{am}} \tag{24}$$

α	$C_{M,0y}$	$C_{M,ca}$
-12°	1.115	0.034
-6°	0.545	0.040
0°	-0.026	0.045
6°	-0.596	0.050
12°	-1.166	0.055

Table 8: Pitching moment coefficients.

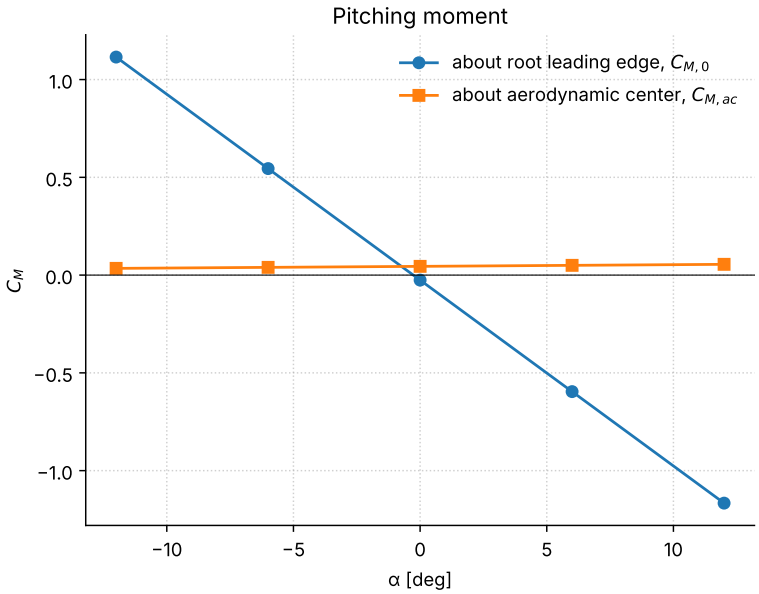


Figure 7: Pitching moment coefficients versus angle of attack.

About the origin the moment varies strongly with α. About the aerodynamic center it is small and nearly constant: it drifts from 0.034 to 0.055 over the whole range, so x_{ca} is correctly located. The small drift, $\mathrm{d}C_{M,ca}/\mathrm{d}C_L \approx 0.009$, means that the exact aerodynamic center of the lattice lies about 2.5 mm ahead of the quarter-MAC estimate. The positive $C_{M,ca}$ comes from the washout of the swept-back tips, which carry negative lift behind the aerodynamic center. With a single chordwise panel all the load acts on the quarter-chord line, so the nose-down moment of the cambered airfoil itself is not represented. Longitudinal stability itself cannot be assessed from these values alone: it requires the center of gravity and the rest of the aircraft (fuselage, tail), which are outside this study.

9. Validation and Mesh Sensitivity

9.1. Comparison with XFLR5

Figure 8 compares the sectional lift with an XFLR5 [3] analysis exported at the same 20 spanwise stations, so the comparison is point by point.

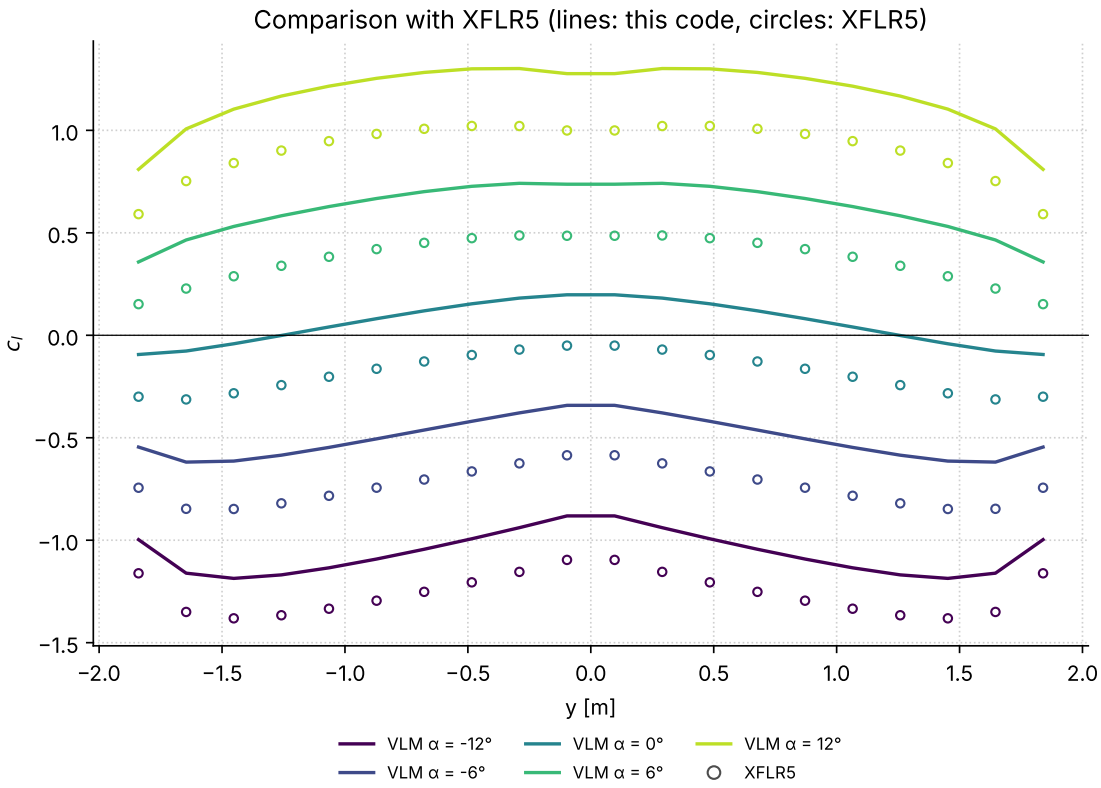


Figure 8: Sectional lift: this code (lines) and XFLR5 (circles) at the same stations.

- **Shape.** Both models give the same distribution shape, including the dip at the plane of symmetry caused by the sweep ($\Lambda_{1/4} = 12°$).
- **Lift slope.** The spacing between curves agrees: integrating the XFLR5 data gives $C_{L\alpha} = 0.0909\,\mathrm{deg}^{-1}$, within 2.6 % of the $0.0932\,\mathrm{deg}^{-1}$ of this code.
- **Absolute level.** There is a near-constant offset of 0.20 to 0.27 in c_l between the two, so the zero-lift angles differ. A likely cause is the camber model: this code applies a single camber slope at $3c/4$ (Equation (1)), while XFLR5 panels the actual mean line.

The agreement therefore validates the slope and the shape of the distribution, not the absolute lift at a given angle.

9.2. Mesh sensitivity

Table 9 and Figure 9 show C_L at $\alpha = 6°$ as the number of spanwise panels (whole wing) increases.

Panels (whole wing)	C_L	Solve time [ms]
4	0.6394	0.2
10	0.6336	0.2
20	0.6277	0.2
40	0.6232	0.4
80	0.6207	1.1
120	0.6199	2.4
160	0.6194	4.4
320	0.6188	14.5

Table 9: Mesh sensitivity at $\alpha = 6°$. Times are machine-dependent.

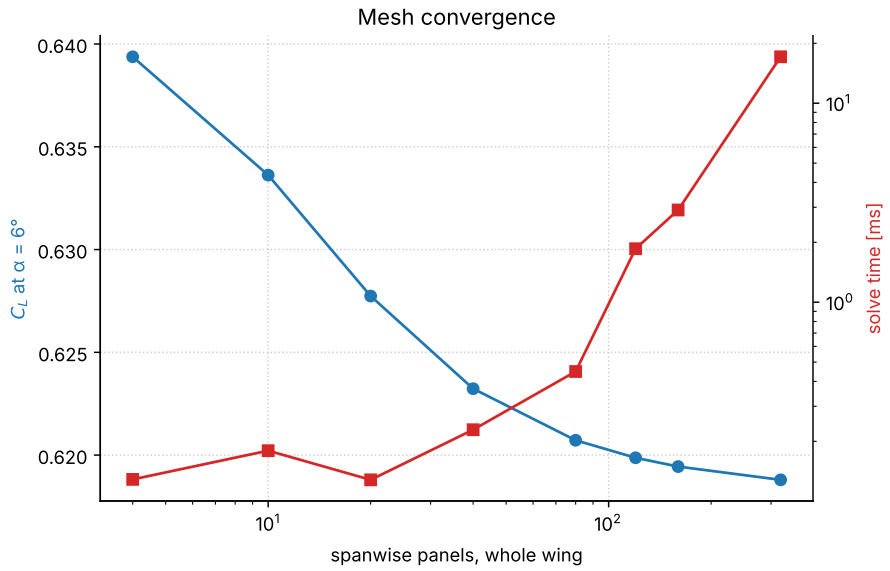


Figure 9: Convergence of C_L and computational cost with the number of panels.

C_L decreases monotonically and changes by less than 0.5 % from 80 panels onward, which is the recommended resolution for accurate results. The cost grows polynomially with the number of panels N (assembly of the $N \times N$ influence matrix and a dense solve), not exponentially, and stays in milliseconds even at 320 panels. The results of Tables 3 to 8 use the 20 panels of the statement; at 80 panels C_L is about 1.1 % lower.

10. Optimization of Stall Characteristics

The objective of this section is to redesign the baseline wing so that the loss of lift begins within 40 % of the semispan (with $y = 0$ at the root) and the sectional lift falls off outboard of that point with a gradient of at least $\mathrm{d}c_l/\mathrm{d}y = -0.2\,\mathrm{m}^{-1}$.

Methodological note. Under the hypothesis of a linear potential regime, stall onset is taken as the station of maximum sectional lift $c_l(y)$ at a high angle of attack ($\alpha = 15°$). It is assumed that the airfoil has a maximum lift coefficient $c_{l,\max}$ that is uniform along the span, so the lift peak marks the section where the boundary layer would separate first in real flight. This is a modelling assumption: the VLM itself does not represent separation.

Baseline. At $\alpha = 15^\circ$ the baseline lift peak is already at 25 % of the semispan, but the gradient outboard of it is only $-0.069\,\mathrm{m}^{-1}$, so the objective is not met.

Search. The code evaluates every combination of aspect ratio (15 to 4), quarter-chord sweep (0 to 24°), taper ratio (0.1 to 1), tip twist (0 to −9°) and airfoil (NACA 0012, 2412, 3217, 4417), about 440 000 designs. The wing area is held at $1\,\mathrm{m}^2$ for every design, so the aspect ratio is exact. The feasible designs are ranked by their normalized distance from the baseline planform:

$$J = \sqrt{\left(\frac{\Delta A}{A_0}\right)^2 + \left(\frac{\Delta \Lambda_{1/4}}{\Lambda_{1/4,0}}\right)^2 + \left(\frac{\Delta \lambda}{\lambda_0}\right)^2 + \left(\frac{\Delta \varepsilon}{\varepsilon_0}\right)^2} \tag{25}$$

Of all the designs, 1 529 meet both requirements. The five closest to the baseline are listed in Table 10.

	Airfoil	A	$\Lambda_{1/4}$	λ	ε	Peak at	$\mathrm{d}c_l/\mathrm{d}y$ [m^{-1}]	J
Baseline	3217	15.0	12°	0.65	−5°	25 %	−0.069	–
1	0012	11.5	3°	1.00	−8°	5 %	−0.200	1.127
2	0012	13.5	5°	1.00	−9°	5 %	−0.200	1.133
3	0012	11.0	3°	1.00	−8°	5 %	−0.201	1.135
4	0012	13.0	5°	1.00	−9°	5 %	−0.201	1.137
5	0012	12.5	5°	1.00	−9°	5 %	−0.201	1.141

Table 10: Baseline and the five feasible designs closest to it ($\alpha = 15^\circ$, 20 panels).

The best design is a NACA 0012 wing with $A = 11.5$, $\Lambda_{1/4} = 3^\circ$, $\lambda = 1.0$ and $\varepsilon = -8^\circ$ (span 3.391 m, constant chord 0.295 m). Its lift peak is at the innermost panel (5 % of the semispan) and the gradient is $\mathrm{d}c_l/\mathrm{d}y = -0.200\,\mathrm{m}^{-1}$ (Figures 10 and 11).

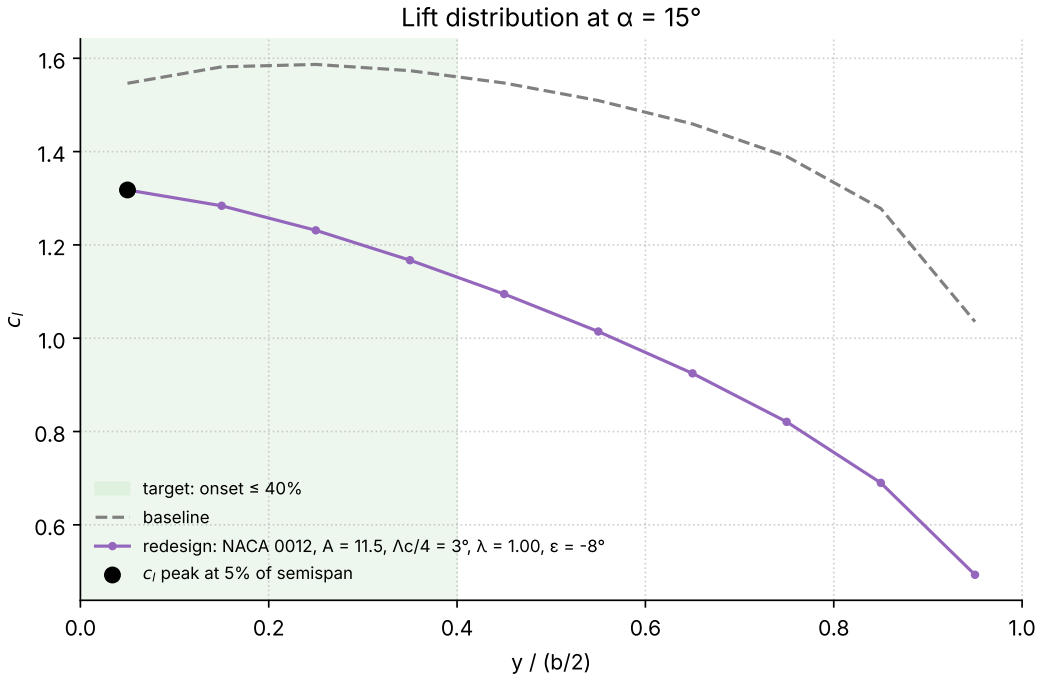


Figure 10: Lift distribution at $\alpha = 15^\circ$ of the baseline and the redesigned wing, with the stall point.

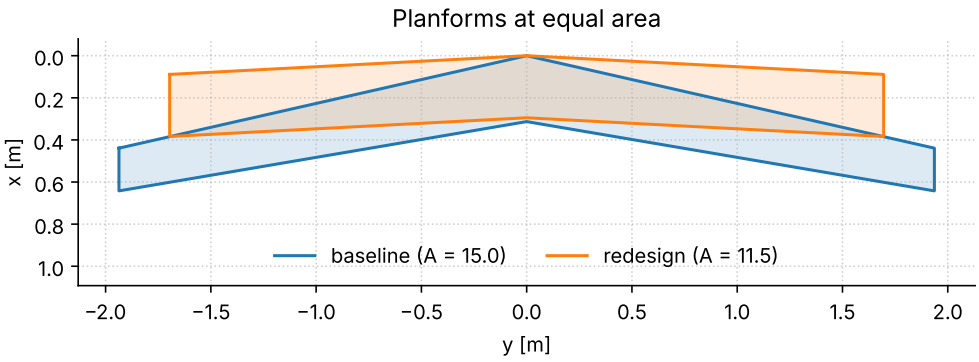


Figure 11: Baseline and redesigned planforms at equal area.

Role of each parameter.

- **Twist** is the most effective parameter. More washout reduces the local angle of attack at the tips and moves the lift peak towards the root. The penalty appears in cruise: the redesigned wing has a sectional lift of -0.39 at the tip at $\alpha = 0$, and the departure from the optimum load raises induced drag and bending loads.
- **Sweep** moves load outboard, so the search reduces it from 12° to 3°.
- **Taper ratio.** A rectangular wing ($\lambda = 1$) has more chord at the tips and carries relatively less load there, which also favours root stall. It moves away from the elliptical distribution and loses efficiency.
- **Aspect ratio** has a secondary effect on where the stall starts, but a lower aspect ratio raises induced drag in cruise.
- **Airfoil.** In this model the airfoil enters only through its camber slope, which shifts the whole distribution without changing its shape. All top designs use the symmetric NACA 0012 because less camber helps meet the gradient threshold at 15°. The real effect of an airfoil change on stall, through $c_{l,\max}$ and thickness, is not captured; in practice a thicker root section would be chosen to make the root stall gently and give warning.

Cost of the redesign. At a cruise lift coefficient $C_L = 0.5$, the redesigned wing needs $\alpha = 9.3^\circ$ instead of 4.6°, and its induced drag rises from $C_{Di} = 0.0053$ to 0.0072 (+36 %; span efficiency 0.96 versus 1.00).

11. Conclusions

A numerical model based on the vortex lattice method has been developed and applied to the aerodynamic characterization of a swept, tapered and twisted wing. The geometric parameters were defined, including the mean aerodynamic chord and the location of the aerodynamic center. The sectional and global lift were computed and split into basic and additional components, which separates the effect of camber and twist from that of incidence. The induced drag was obtained in the Trefftz plane, and the pitching moment about the aerodynamic center was found to be small and nearly independent of the angle of attack.

The results were compared with the open-source software XFLR5, which confirms the lift slope (within 2.6 %) and the shape of the distribution, and a mesh study shows convergence to within 0.5 % from 80 panels onward. Finally, a search over aspect ratio, sweep, taper, twist and airfoil produced designs that move the onset of stall to the wing root with the required lift gradient, mainly through more washout, less sweep and a rectangular planform, at the price of roughly a third more induced drag in cruise.

Limitations: the model is linear and inviscid, uses one chordwise panel, and predicts the location of the stall only through the lift-peak criterion, not through boundary-layer separation.

12. Appendix

Code. The complete code, tests and figures are available on GitHub:
`https://github.com/CPesquero/vlm-wing-aerodynamics`.
Every table and figure of this report can be reproduced with the command `vlmwing`.

Revision note. This revised edition (October 2026) corrects the first version of the report:

- The XFLR5 comparison and the stall optimization had been computed with a 20° leading-edge sweep instead of the baseline $\Lambda_{1/4} = 12°$ ($\Lambda_{LE} = 12.78°$).
- The optimization did not keep the wing area constant, so the reported aspect ratio ($A = 7.24$) was not the real one. The optimization results (Table 10) have been recomputed.
- Corrected statements: the leading-edge sweep formula (Equation (7)); XFLR5 does reproduce the root dip; computing cost grows polynomially, not exponentially; a constant $C_{M,ca}$ alone does not establish longitudinal stability; XFLR5 is open-source, not commercial.
- Minor corrections to rounded values in Tables 3 and 5, and to figure, table and equation numbering.

Sections 1 to 8 and the mesh-sensitivity values are otherwise unchanged.

References

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